Stroma World — Reality Check Suite v1.0

> **What this is:** Five integration tests that the fully assembled simulation must pass before any EA evolution begins and before Cycle 2 (immune evasion) is scoped. **How this differs from Reality Anchors:** The 10 Reality Anchors test individual subsystems and wiring (does ECM block diffusion? does SHH inhibition worsen outcomes?). These 5 Reality Checks test whether the *whole system* produces emergent behavior that matches PDAC clinical reality when all subsystems run together. An anchor can pass in isolation while a check fails because checks depend on multi-module interactions that only manifest in full-system runs. **Gate rule:** All 5 must pass. If any check fails, do not proceed. Debug, fix, re-run all 5.

Check 1: The Untreated Natural History

The Question

If you place a PDAC tumor in a field of quiescent stellate cells and do absolutely nothing — no drug, no intervention, no immune cells — does the simulation produce something that looks like untreated PDAC over a timescale of weeks?

Setup

- Seed a small cluster of tumor cells (50–100) with canonical PDAC genotype at the domain center.
- Surround with quiescent PSCs (200–500) distributed in the peri-tumoral zone.
- All microenvironment fields at initial conditions: O_2 at boundary value, TGF-β = 0, SHH = 0, Drug = 0, ECM = 0.
- No interventions. No drug. Run for 4–6 weeks simulated time.
- Minimum 5 replicate runs with different random seeds.

Expected Outcome — Six Things That Must All Happen, in Roughly This Order

Week 1: Tumor cells begin secreting TGF-β and SHH. Nearest PSCs sense signals and activate (ACTA2 → ON). First ECM deposits appear adjacent to tumor. Tumor cell count grows steadily. No hypoxia yet — the tumor is small and ECM is thin.

Week 2: CAF population expands through both activation of remaining PSCs and CAF proliferation. ECM density rises in a ring around the tumor. The ring is concentric, densest at the tumor-stroma interface, thinning outward. O_2 gradient begins forming — slight dip at tumor centroid. First tumor cells at the periphery may begin EMT if TGF-β is concentrated enough at the boundary.

Weeks 3–4: Barrier is now substantial. O_2 at centroid is measurably lower than at periphery. HIF1A activates in central tumor cells. Hypoxia-driven TGF-β amplification kicks in — the positive feedback loop accelerates barrier construction. EMT is clearly visible at the tumor-stroma interface: peripheral cells are mesenchymal (ZEB1=ON), core cells remain epithelial. Tumor-to-stroma ratio approaches a PDAC-like range (more stroma than tumor by area).

Weeks 4–6: System approaches a dynamic steady state. Barrier is mature. Tumor growth slows at the boundary (mechanical confinement) but continues in pockets where ECM is thinner. Central hypoxia is pronounced. The spatial architecture matches PDAC histology at a qualitative level: dense desmoplastic ring, hypoxic tumor core, mesenchymal invasive front, epithelial interior.

Pass Criteria (all must be met, across all replicates)

ID	Criterion
C1.1	Barrier self-assembles: mean peri-tumoral ECM at endpoint > 0.5 (on the 0–1 scale). Stroma is dense, not wispy.
C1.2	Spatial gradient exists: ECM density at tumor boundary > ECM density at domain edge by at least a factor of 3. The barrier has structure, not uniform haze.
C1.3	Hypoxia develops: O_2 at tumor centroid at endpoint < O_2 at tumor periphery at endpoint. A gradient exists.
C1.4	EMT is spatial: emt_fraction_periphery > emt_fraction_core in at least 4 of 5 replicates.

ID	Criterion
C1.5	Tumor-to-stroma ratio is PDAC-like: tumor cells constitute less than 50% of total cell area at endpoint. If the tumor dominates the domain with minimal stroma, the paracrine loop is too weak.
C1.6	The system does not blow up: no field exceeds its boundary (ECM ≤ 1.0, O_2 ≤ boundary value, no negative concentrations). No cell populations go to infinity or zero unexpectedly.

Failure Interpretation

Failure	Most likely cause	First place to look
No barrier forms (C1.1 fails)	Paracrine loop broken	Check Knobs 1a/1b → TGF-β/SHH secretion rates. Check PSC activation threshold. Check edges E06, E07, E10.
Barrier forms but is spatially uniform (C1.2 fails)	TGF-β/SHH diffusing too fast or ECM not impeding signal diffusion	Check diffusion coefficients for TGF-β and SHH. Check Rule 36 (ECM traps signals). Reduce diffusion rates or increase ECM→diffusion coupling.
No hypoxia (C1.3 fails)	ECM not impeding O_2 diffusion, or domain is too small	Check ECM→O_2 diffusion coupling (edge E20). If coupling is correct, domain may be too small for a meaningful O_2 gradient to develop — increase domain size or decrease boundary O_2.
EMT is everywhere or nowhere (C1.4 fails)	Threshold too low (everywhere) or too high (nowhere)	Adjust Knob 5a. If EMT is uniform, also check whether TGF-β is flooding the domain rather than concentrating at the interface.
Too much tumor, too little stroma (C1.5 fails)	Tumor proliferation outpacing CAF activation + ECM production	Either slow tumor (increase Knob 3a denominator mapping) or accelerate CAF activation (lower activation threshold) or increase ECM production rate. The ratio is a calibration target — adjust until it matches PDAC histology.

Failure	Most likely cause	First place to look
Numerical explosion (C1.6 fails)	Time step too large, PDE instability, or runaway positive feedback without damping	Check CFL condition for PDE solver. Check that ECM density is clamped at 1.0 (Rule 63). Check that the TGF-β positive feedback loop (E12) has natural saturation (signal decay, PSC depletion).

Why This Check Gates Phase 2

If the untreated natural history is wrong, every intervention test built on it is meaningless. You cannot test whether a drug penetrates the barrier if the barrier doesn't form correctly. You cannot test immune exclusion (Cycle 2) if the stroma architecture doesn't match PDAC. This check validates the stage before any actors perform on it.

Check 2: Standard-of-Care Chemotherapy Fails Realistically

The Question

When you apply gemcitabine-equivalent drug at maximum tolerated dose against a mature barrier, does the simulation reproduce the clinical reality: modest tumor shrinkage, incomplete response, rapid regrowth after withdrawal?

Setup

- Run Check 1 to produce a mature barrier (4 weeks untreated).
- At week 4, apply drug at the domain boundary at maximum tolerated concentration (the ceiling from Rule 59). Continuous exposure for 2 weeks simulated time.
- At week 6, withdraw drug. Continue simulation for 2 additional weeks (observation of regrowth).
- Total simulation: 8 weeks. Drug window: weeks 4–6.
- 5 replicates.

Expected Outcome

During treatment (weeks 4–6):

- Drug penetrates partially into the stroma. Drug concentration at tumor centroid is low — a fraction of the boundary concentration.
- Tumor cells nearest the drug source (periphery, thinnest barrier) begin dying. Tumor cell count decreases, but slowly.
- Cells deeper in the tumor (behind dense ECM) experience little or no drug. They survive.
- Over the 2-week treatment, ABCB1 activates in drug-exposed cells that survived initial exposure. fraction_abcb1_active rises from 0 toward a plateau.
- CAFs are unaffected. ECM density does not decrease. The barrier holds.

After withdrawal (weeks 6–8):

- Drug washes out. Intracellular drug decays.
- Surviving tumor cells — concentrated in the spatial sanctuary behind dense ECM — resume proliferation.
- These survivors resume TGF-β and SHH secretion. CAFs (which never stopped) continue producing ECM.
- Tumor cell count begins climbing back toward pre-treatment levels.
- By week 8, the tumor is visibly regrowing and the barrier is intact or rebuilding where it was partially thinned by MMP2 at the invasive front.

Pass Criteria

ID	Criterion
C2.1	Drug does not eradicate the tumor. total_tumor_cells at week 6 (end of treatment) is greater than zero. Some cells survive.

ID	Criterion
C2.2	Drug does reduce the tumor. total_tumor_cells at week 6 < total_tumor_cells at week 4 (start of treatment). Treatment does something — it's just not enough.
C2.3	Tumor regrows after withdrawal. total_tumor_cells at week 8 > total_tumor_cells at week 6. The sanctuary population rebounds.
C2.4	Barrier persists through treatment. mean_peritumoral_ecm at week 6 ≥ 80% of mean_peritumoral_ecm at week 4. The drug does not meaningfully erode the stroma.
C2.5	Survivors are spatially concentrated. survivor_mean_ecm at week 6 > mean ECM at dead cell locations. Survivors are behind denser barrier, not randomly distributed.
C2.6	Resistance emerges. fraction_abcb1_active at week 6 > fraction_abcb1_active at week 4 (which was 0). Drug exposure induced efflux in surviving cells.

Failure Interpretation

Failure	Most likely cause	First place to look
Drug eradicates tumor (C2.1 fails)	Drug penetrating too easily or death resistance too low	Check ECM→drug diffusion coupling (edge E21). Check Knob 4 (apoptosis resistance). If drug easily reaches the centroid, the barrier isn't blocking diffusion.
Drug has no effect (C2.2 fails)	Drug not penetrating at all or kill signal not reaching Module 4	Check drug diffusion coefficient (may be too low — drug doesn't move). Check M7→M4 kill signal path (edges E24). Check that the drug PDE is actually being solved.

Failure	Most likely cause	First place to look
No regrowth (C2.3 fails)	Survivors lost proliferative capacity or paracrine drive	Check Rule 4 — do daughter cells inherit correct genotype? Check that surviving cells' secretion rates are intact. If survivors have aberrant state, the death/survival selection may be corrupting cell state.
Barrier collapses during treatment (C2.4 fails)	Drug accidentally killing CAFs, or MMP2 degradation outpacing production	Check Rule 31 (CAFs are drug-indifferent). If CAFs are dying, the drug response module is incorrectly targeting them. If MMP2 is the cause, check whether EMT is too aggressive (too many MMP2+ cells).
Survivors are randomly distributed (C2.5 fails)	Drug penetrating uniformly despite ECM, or death probability not drug-concentration-dependent	Check that apoptosis rate is a function of intracellular drug (not a flat probability). Check that ECM actually creates a drug gradient (spatial heterogeneity of drug exposure).
No resistance (C2.6 fails)	ABCB1 induction pathway disconnected	Check M7 internal logic: NRF2 activation → timer → ABCB1. Check Knob 7a (delay). Check edge E25.

Why This Check Gates Phase 2

PDAC's response to standard chemotherapy is the most well-documented clinical behavior. If the simulation can't reproduce "modest response, incomplete kill, rapid relapse," it will produce fantasy results when the EA tries to optimize treatment. Every EA-discovered strategy will be benchmarked against this standard-of-care baseline. If the baseline is wrong, the benchmarks are meaningless.

Check 3: The Vismodegib Paradox Emerges Spontaneously

The Question

If the EA (or a test scenario) tries the "obvious" strategy of removing stroma, does the simulation

correctly show that this makes things worse without concurrent cytotoxicity?

Setup

Three parallel arms, all starting from the same mature barrier state (week 4 of an untreated run):

- **Arm A — Control:** No intervention. Run 4 additional weeks.
- **Arm B — SHH inhibition only:** Set shh_secretion_rate = 0 at week 4. No drug. Run 4 additional weeks.
- **Arm C — SHH inhibition + drug:** Set shh_secretion_rate = 0 AND apply drug at max tolerated dose at week 4. Run 4 additional weeks.

5 replicates per arm (15 total runs).

Expected Outcome

Arm A (control): Tumor grows slowly, constrained by barrier. Barrier is stable. Tumor cell count increases moderately.

Arm B (SHH inhibition, no drug): GLI1 turns off in all CAFs. ECM production rate drops (GLI1 boost removed, but ACTA2/TGF-β arm still active). Barrier thins partially. Mechanical confinement decreases. Tumor, no longer fully caged, expands into the freed space. Tumor cell count at endpoint is HIGHER than Arm A. Tumor radius is LARGER. The "treatment" made things worse.

Arm C (SHH inhibition + drug): Same stroma thinning as Arm B, but now drug is present. Thinner stroma means drug penetrates deeper. Drug kills more tumor cells than in Check 2 (where barrier was intact). Tumor cell count at endpoint is LOWER than Arm A. The combination works — stroma thinning is beneficial *only when paired with a killing agent*.

Pass Criteria

ID	Criterion
C3.1	Arm B is worse than Arm A. total_tumor_cells at endpoint: B > A. SHH inhibition alone is

ID	Criterion
	counterproductive.
C3.2	Arm B tumors are bigger. tumor_radius at endpoint: B > A. Physical expansion into freed space.
C3.3	Arm B barrier is thinner. mean_peritumoral_ecm at endpoint: B < A. Confirms the intervention actually changed the stroma.
C3.4	Arm C is better than Arm A. total_tumor_cells at endpoint: C < A. The combination works.
C3.5	Arm C is better than Check 2 (drug alone against intact barrier). total_tumor_cells at endpoint of Arm C < total_tumor_cells at endpoint of Check 2 treatment (week 6 of Check 2). Stroma thinning synergizes with drug.
C3.6	The rank order is unambiguous: C < A < B for tumor cell count. Combination best, control middle, SHH inhibition alone worst.

Failure Interpretation

Failure	Most likely cause	First place to look
Arm B is better or equal to Arm A (C3.1 fails)	Mechanical confinement too weak — removing stroma doesn't release the tumor	Check Module 8, edge E26 (solid stress → contact inhibition). Check Rule 37. If the tumor wasn't mechanically constrained in Arm A, removing the constraint has no effect. Increase Knob 8 or collagen-mediated stiffness.
Arm B barrier unchanged (C3.3 fails)	SHH pathway not connected to ECM production	Check edges E07, E11. Check that GLI1 is actually turning off when SHH=0. Check that GLI1=OFF reduces ECM production rate in Module 6.
Arm C not better than Arm A (C3.4	Stroma thinning didn't improve drug penetration, or	Check ECM→drug diffusion coupling. If barrier thinned but drug still doesn't penetrate, the diffusion coefficient

Failure	Most likely cause	First place to look
fails)	drug is too weak even with better access	mapping may not be sensitive enough to ECM changes.
Rank order scrambled (C3.6 fails)	Multiple interacting failures	Run the failure interpretations for whichever pairwise comparison broke. If A < B is correct but C > A, focus on drug efficacy. If B < A but C < A, focus on mechanical confinement.

Why This Check Gates Phase 2

The vismodegib paradox is the most counterintuitive fact about PDAC stroma biology. If the simulation reproduces it spontaneously — without being explicitly told "SHH inhibition should worsen things" — it means the model has captured the dual nature of stroma (shield AND cage) through emergent behavior from Rules 37–39. Any EA running on this model will face the same paradox and must find solutions that navigate it. If the paradox doesn't emerge, the EA will find naive stroma-removal strategies that would fail catastrophically in patients.

Check 4: The EA Can Distinguish Good From Bad Strategies

The Question

Does the fitness function correctly rank a set of known strategies from best to worst — specifically, does it rank clinically validated combinations above clinically failed monotherapies?

Setup

Run four intervention strategies against the same mature barrier, each for the same treatment duration:

- **Strategy 1 — Drug only (standard of care):** Drug at max dose, no stroma intervention. (Clinically: modest response.)

- **Strategy 2 — SHH inhibition only:** SHH blocked, no drug. (Clinically: failure, tumors accelerated.)
- **Strategy 3 — ECM degradation only:** Global ECM degradation (simulating hyaluronidase), no drug. (Clinically: failure as monotherapy.)
- **Strategy 4 — ECM degradation + Drug:** ECM degradation simultaneous with max dose drug. (Clinically: improved drug delivery, better than drug alone.)

Compute the fitness score for each strategy using the fitness function from the EA spec.

5 replicates per strategy (20 total runs).

Expected Outcome

The fitness function should rank the strategies:

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BEST:   Strategy 4 (ECM degradation + Drug)
SECOND: Strategy 1 (Drug only)
THIRD:  Strategy 3 (ECM degradation only)
WORST:  Strategy 2 (SHH inhibition only)
```

This ranking matches clinical evidence: combination therapy > monotherapy > barrier removal without killing.

Pass Criteria

ID	Criterion
C4.1	Fitness(Strategy 4) > Fitness(Strategy 1). Combination beats standard of care.
C4.2	Fitness(Strategy 1) > Fitness(Strategy 2). Standard of care beats SHH inhibition alone.
C4.3	Fitness(Strategy 1) > Fitness(Strategy 3). Standard of care beats ECM degradation alone.
C4.4	Fitness(Strategy 2) is the lowest or near-lowest. SHH inhibition alone is actively harmful.

ID	Criterion
C4.5	The fitness function's ranking is consistent across all 5 replicates. The ordering is not a stochastic accident.

Failure Interpretation

Failure	Most likely cause	First place to look
Strategy 4 not ranked best (C4.1 fails)	Fitness function doesn't reward drug penetration improvement, or ECM degradation doesn't improve drug access	Check fitness function formulation — does it account for tumor kill, not just tumor count? Check ECM→drug coupling.
Strategy 2 not ranked worst (C4.2/C4.4 fails)	Fitness function doesn't penalize tumor growth, or confinement release effect is too weak	Check that fitness function includes tumor burden as a penalty. Check mechanical confinement (Module 8).
Strategy 3 ranked above Strategy 1 (C4.3 fails)	ECM degradation alone is providing unrealistic benefit — possibly because barrier removal is permanent and not rebuilding	Check Rule 29 (ECM turnover) and edge E10 (CAF ECM rebuilding). If ECM doesn't regrow after degradation, the model is too optimistic about degradation monotherapy.
Ranking is inconsistent across replicates (C4.5 fails)	Too much stochastic noise relative to inter-strategy differences	Increase replicate count. If ranking is still inconsistent, the strategies may be too similar in effect — check whether parameter values are creating a regime where the interventions barely matter.

Why This Check Gates Phase 2

This check validates the fitness function, not the biology. The biology was validated by Checks 1–3. Check 4 asks: given correct biology, does the scoring system correctly reward good strategies and punish

bad ones? If the fitness function can't distinguish a known-good from a known-bad strategy, it will mislead the EA — the optimizer will converge on strategies that score well but are biologically nonsensical. This is the last validation before the EA is trusted to search autonomously.

Check 5: The Sanctuary-Regrowth Cycle Completes

The Question

After the best available treatment (Strategy 4 from Check 4), does a residual tumor population survive in a spatial sanctuary, and does that population successfully rebuild the tumor and its barrier when treatment stops? This is the full disease recurrence cycle that makes PDAC lethal.

Setup

- Start from mature barrier (week 4 of untreated run).
- Apply Strategy 4 (ECM degradation + drug) for 2 weeks (weeks 4–6).
- Withdraw all intervention at week 6.
- Continue simulation for 4 weeks post-treatment (weeks 6–10).
- 5 replicates.

Expected Outcome — The Five Phases of Recurrence

Phase 1 — Treatment response (weeks 4–5): ECM degrades. Drug penetrates deeper than in Check 2. Tumor cells die — more than with drug alone. Tumor shrinks meaningfully.

Phase 2 — Residual survival (week 5–6): Despite aggressive treatment, cells in the densest remaining ECM (deepest interior) survive. These cells were shielded by the residual barrier and have activated ABCB1 from partial drug exposure. `total_tumor_cells` bottoms out at a nonzero floor — partial response, not complete response.

Phase 3 — Early regrowth (weeks 6–7): Treatment withdrawn. Drug washes out. ECM begins rebuilding from persisting CAFs (which were never drug-targeted). Surviving tumor cells resume

proliferation. Cell count begins climbing. The cells are initially concentrated in the sanctuary zone.

Phase 4 — Barrier reconstruction (weeks 7–9): Growing tumor cells resume TGF-β and SHH secretion. Remaining PSCs (if any) activate. Existing CAFs ramp production. ECM density at the tumor-stroma interface rises back toward pre-treatment levels. The barrier is rebuilding.

Phase 5 — Recurrence established (weeks 9–10): Tumor cell count approaches or exceeds pre-treatment levels. Barrier density approaches pre-treatment levels. The spatial architecture (hypoxic core, mesenchymal front, dense stroma ring) re-establishes. The system has returned to a state qualitatively similar to the starting point — the treatment delayed the disease but did not alter its fundamental trajectory.

Pass Criteria

ID	Criterion
C5.1	Treatment reduces tumor. total_tumor_cells at week 6 (treatment end) < total_tumor_cells at week 4 (treatment start). The best available strategy does have an effect.
C5.2	Treatment does not eradicate. total_tumor_cells at week 6 > 0. A residual population survives.
C5.3	Survivors are spatially concentrated. survivor_mean_ecm at week 6 > mean ECM at dead cell locations. The sanctuary is real.
C5.4	Tumor regrows after withdrawal. total_tumor_cells at week 10 > total_tumor_cells at week 6. The sanctuary population rebounds.
C5.5	Barrier rebuilds after withdrawal. mean_peritumoral_ecm at week 10 > mean_peritumoral_ecm at week 6. Stroma returns.
C5.6	The regrowing tumor re-establishes spatial architecture. At week 10: O_2 at centroid < O_2 at periphery (hypoxia returns), AND emt_fraction_periphery > emt_fraction_core (EMT spatial pattern returns). The recurrence isn't just cell count recovery — it's full phenotypic and architectural reconstitution.

ID	Criterion
C5.7	Regrowth population contains drug-hardened cells. fraction_abcb1_active in the regrowing population at week 8 > 0. Some survivors carry activated efflux from the treatment phase, meaning a second treatment cycle would face partial resistance from the start.

Failure Interpretation

Failure	Most likely cause	First place to look
Treatment eradicates tumor (C5.2 fails)	Drug + degradation too effective — barrier physics or resistance mechanisms too weak	Check that ECM degradation is partial, not total (some ECM should survive even after degradation). Check Knob 4, Knob 7b. If all tumor cells die, resistance is too weak for this regimen.
No spatial concentration of survivors (C5.3 fails)	Drug penetrating uniformly despite residual ECM	Check ECM→drug diffusion coupling. If ECM degradation is too thorough, all voxels reach near-zero ECM and there's no spatial gradient left to create a sanctuary. Reduce degradation intensity.
No regrowth (C5.4 fails)	Surviving cells can't divide or can't secrete	Check that surviving cells have intact genotype (Rule 4). Check that post-treatment cells resume secretion (Rule 13). If survivors are in growth arrest from mechanical pressure, check whether ECM rebuilt enough to compress them — but CAF-produced ECM shouldn't be compressing the tumor immediately after degradation.
No barrier rebuild (C5.5 fails)	CAFs depleted or disconnected from production	Check Rule 31 (CAFs survive drug). Check that ACTA2=ON CAFs still have active ecm_production_rate. If CAFs are alive but not producing, check Module 6 and edges E10, E12.
No architecture reconstitution	Regrowth is happening but without the spatial	Check whether the regrowing tumor is too small at week 10 to generate a meaningful O_2 gradient or TGF-β gradient. If so, extend

Failure	Most likely cause	First place to look
(C5.6 fails)	organization	the post-treatment observation period. If the tumor is large enough but lacks architecture, check that the feedback loops (Loops 1 and 2 from Edge List) are closing correctly in the post-treatment context.
No resistance carryover (C5.7 fails)	ABCB1 decayed too fast after drug withdrawal	Check the NRF2/ABCB1 decay rate (Rule 11). If decay is too fast, drug-adapted cells revert to naive before regrowth is measurable. Slow the decay constant.

Why This Check Gates Phase 2

This is the acid test. It simulates the full clinical trajectory of PDAC: growth → barrier formation → treatment → partial response → sanctuary survival → regrowth → recurrence. If the simulation reproduces this complete cycle, it has captured the essential treatment-resistance dynamic of PDAC. The EA will then search for interventions that *break* this cycle — strategies that either eliminate the sanctuary, prevent regrowth, or stop barrier reconstruction. If the cycle doesn't complete in simulation, the EA has nothing meaningful to optimize against.

This check is also the bridge to Cycle 2: the regrowing, barrier-rebuilt tumor at week 10 is exactly the starting state for immune evasion modeling. Cycle 2 asks "can immune cells breach this barrier and kill this tumor?" If the barrier at week 10 doesn't look like real PDAC stroma, the immune evasion cycle will be fighting the wrong enemy.

Suite Execution Protocol

```
Step 1: Run Check 1 (Natural History)
  └─ PASS → proceed
  └─ FAIL → debug, fix, re-run Check 1 until it passes

Step 2: Run Check 2 (Standard of Care Failure)
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  └─ PASS → proceed
  └─ FAIL → debug, fix, re-run Checks 1–2

Step 3: Run Check 3 (Vismodegib Paradox)
  └─ PASS → proceed
  └─ FAIL → debug, fix, re-run Checks 1–3

Step 4: Run Check 4 (Fitness Function Ranking)
  └─ PASS → proceed
  └─ FAIL → likely a fitness function problem, not a biology problem
        (Checks 1–3 already validated biology)
        Fix fitness function, re-run Check 4 only

Step 5: Run Check 5 (Sanctuary-Regrowth Cycle)
  └─ PASS → ✓ PHASE 1 COMPLETE. Model is validated.
        EA evolution may begin.
        Cycle 2 scoping may begin.
  └─ FAIL → debug, fix, re-run Checks 1–5

Total runs: 5 + 5 + 15 + 20 + 5 = 50 minimum
```

Escalation Rule

If any single check fails more than 3 times after debugging, escalate to a full review of the module wiring and edge list. Repeated failures on the same check suggest a structural problem (wrong edge, missing feedback loop, incorrect module execution order) rather than a parameter tuning issue.

What Passing All 5 Checks Means

It means the simulation produces:

1. A self-assembling stroma barrier with correct spatial architecture (Check 1)

2. Realistic treatment failure against the barrier (Check 2)
3. The correct counterintuitive response to naive stroma removal (Check 3)
4. A fitness landscape that matches clinical evidence (Check 4)
5. The full recurrence cycle from sanctuary populations (Check 5)

This is sufficient — not to claim the model is "true," but to claim the model is a *fair opponent* for the EA. The EA will face the same layered defense (Rule 54) that real drugs face. Its discoveries will be constrained by the same biology that constrains real therapy. And its failures will fail for the same reasons real treatments fail.

That is all a minimal stroma world needs to be.

Five checks. Fifty runs. If they all pass, the world is ready. If any fails, the failure interpretation tells you where to look. No check is optional.